

Agentic AI Safety and Alignment

Core Concepts of Agentic AI Safety and Alignment



Steve Buchanan

Cloud Thought Leader

@buchatech | www.buchatech.com



Course Introduction

Core concepts of agentic AI safety and alignment

Aligning AI agents with organizational values



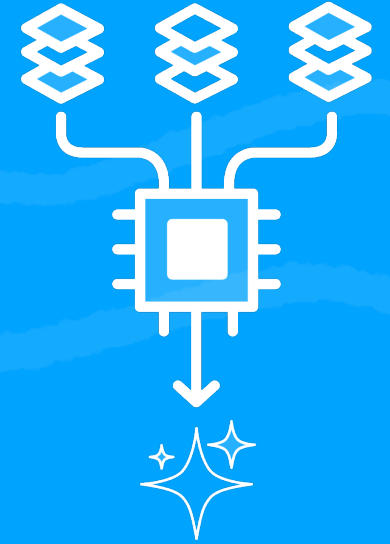


What AI Agents Are



What Are AI Agents?

“An AI Agent is software that uses artificial intelligence to perceive their environment, make decisions, & execute tasks using integrated tools & data sources.”



They think and decide

1. You give the AI Agent a job
2. The AI agent thinks about how to do it
3. Then the AI Agent goes and does the job for you

They use different tools

They can gather input (from systems, databases, sensors, etc.), analyze it & perform tasks

AI agents often combine:

- LLMs (like GPT-4) for reasoning & planning
- Tools or APIs for taking actions (e.g., sending emails, querying DBs),
- Memory for keeping context over time

They work until the job is done

They can perform multi-step processes, react to changing environments, and revise their strategy for iterative or continuous action until the task is done



Common AI Agents



LangChain



Microsoft Copilot



OpenAI GPT



AutoGPT



Google Vertex AI
agent



CrewAI



Common AI Agent Use Cases



A customer support bot that solves complex tickets automatically



A software agent that monitors cloud resources and scales services as needed



A research assistant agent that browses the web, summarizes content, and compiles reports



A sales outreach agent that drafts personalized emails, sends follow-ups, and updates the CRM based on lead responses





Safety and Alignment in Agentic AI



Safety and Alignment in Agentic AI

Safety

AI agents are **autonomous** and often have **access** to **sensitive systems** and **tools**

It's essential to **establish** clear **boundaries** and rules to guide their behavior. **Without safeguards**, agents can be exploited, misused, or behave unpredictably, leading to security risks, compliance issues, or runaway costs

AI administrators **must ensure** these **agents operate safely** within the organization's environments to prevent harm and maintain control



Alignment

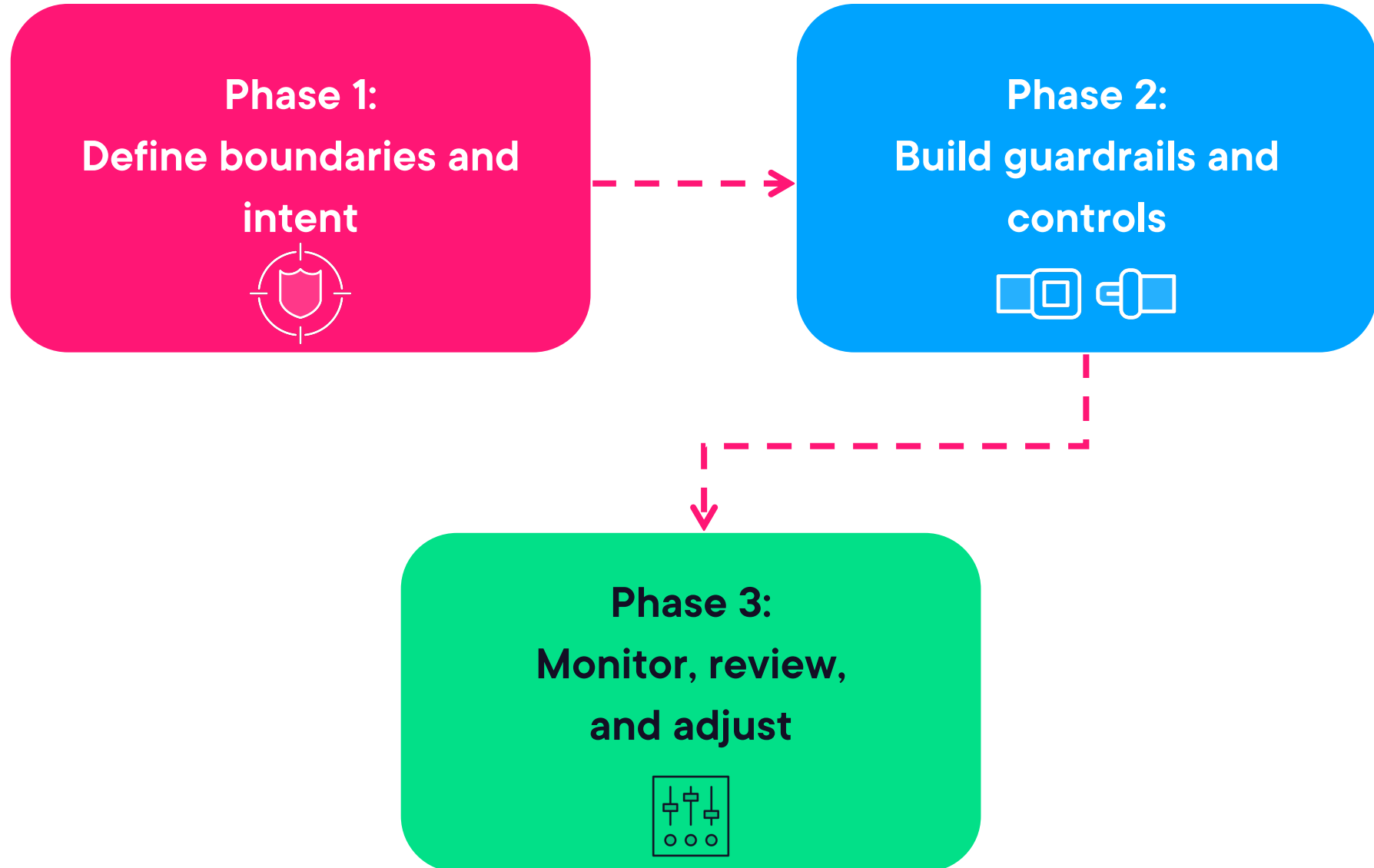
Organizations **operate** with a **mission, vision**, and a clear set of **ethics** and **values**, often outlined in the employee handbook

Employees are expected to follow these principles, which guide how they interact with coworkers, handle customer data, perform their work, and engage with customers

AI agents should **be held to the same standard, aligning their behavior** with these **values** to ensure consistency, trust, and ethical outcomes



Safety and Alignment in an Agentic AI Strategy



Phase 1: Define Boundaries and Intent

Goal: Set clear expectations for what the agent should and should not do

- Identify the agent's role, goals, and responsibilities
- Translate org values, ethics, and mission into behavioral guidelines
- Define hard boundaries like:
 - What tools the agent can access
 - What types of tasks it can perform
 - What types of data it can use or generate
- Use plain-language prompts, config files, or system messages to reinforce this intent across platforms



Why it matters: Without clearly defined intent, agents may misinterpret goals, take shortcuts, or behave in ways that conflict with company values



Phase 2: Build Guardrails and Controls

Goal: Prevent unsafe or unintended behavior before it happens

- Implement safety checks such as:
 - Input validation & prompt filtering
 - Output moderation for harmful or biased content
 - Rate limits or task recursion limits
 - Human-in-the-loop approval for sensitive actions
- Use built-in tools (e.g., Flowise guardrails, LangChain callbacks, OpenAI moderation) or custom logic to enforce constraints

Why it matters: Even well-intentioned agents can be tricked, exploited, or run out of control without technical safeguards in place

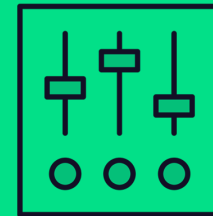


Phase 3: Monitor, Review, and Adjust

Goal: Ensure the agent stays aligned over time

- Set up logging & feedback loops to track agent behavior & decisions
- Periodically review outputs for ethical or operational misalignment
- Adjust prompts, rules, or access based on observed behavior & changing business needs
- Encourage human oversight, especially in high-stakes or customer-facing tasks

Why it matters: Alignment can drift as tasks change, or agents interact with real-world complexity, ongoing review helps catch & correct drift





What AI Agent Runaway Behavior Is



What Is Agent Runaway Behavior?

Agent runaway behavior is when an AI agent gets stuck in an uncontrolled, escalating loop of unintended actions

It's when an agent deviates from its original goals and rapidly waste resources like time, money, or computing power



Agent Runaway Behavior



Endless loops



Over-tasking



Prompt exploitation (i.e. bad actors hijacking agent behavior)



Tool misuse (i.e. unsafe use of external APIs or databases)

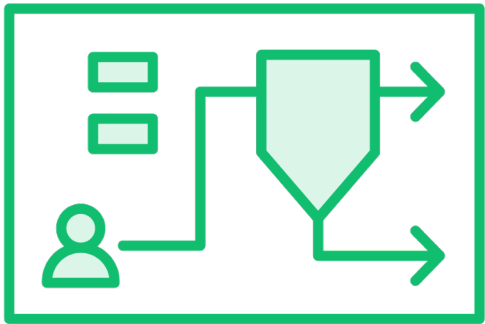




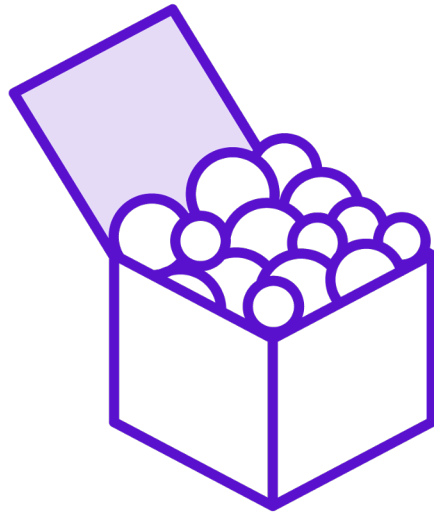
Building Rules and Safety Checks in Agentic AI Systems



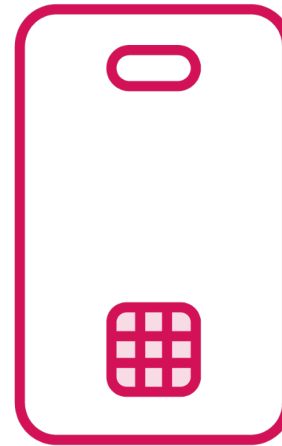
Building Rules and Safety Checks in Agentic AI



Human-in-the-
Loop Escalation



Prompt Injection
Detection



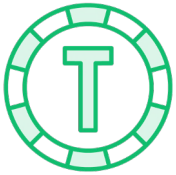
Token & Response
Size Limits



Guardrails via
System Prompts

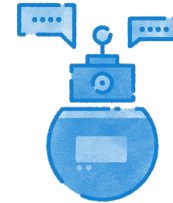


AI Control Methods



Max Tokens: Limits how long the AI's response can be. Prevents long/rambling answers, reduces latency, controls costs.

Use: Max_tokens=200 will cap the response at around 200 tokens (roughly 150–200 words).



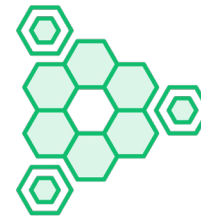
System Prompting: High-level instructions to the model about how it should behave.

Use: "You are a helpful assistant. Keep all responses under 3 paragraphs."



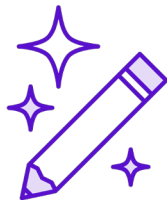
Temperature: Limits how long the AI's response can be.

Use: Lower values (0.5–0.7): More focused, consistent, and predictable. Higher values (0.8–1.0): More diverse/creative, may be less accurate/relevant



Stop Sequences: Tells the model when to stop generating text by using specific phrases or patterns to prevent "hallucinating".

Use: Adding "\nUser:" or "###" to stop the model when it hits a new speaker or section.



Guardrails (Pre/Post): Adds logic before or after the LLM call to guide or validate responses.

Use: Increase safety, adheres to policies, & keeps responses aligned with org goals.

Pre-guardrails: Input checks, context enforcement, or safety filters before the prompt reaches the end user

Post-guardrails: Filters, rule enforcement, or moderation after the LLM responds



Prompt Injection Detection Flow

[User Input]



[Input Monitoring]

What it is: Watch everything the user or external system sends to the agent



[Intent and Context Check]

What it is: The agent checks if the new prompt aligns with its current task and memory/state



[Content Filtering]

What it is: Use predefined filters (regex, keywords) to flag toxic, manipulative, or suspicious input



[Rule Enforcement]

What it is: The agent checks prompt instructions against a set of guardrails or system policies



[Log + Feedback Loop]

What it is: Logs potentially dangerous prompts and feeds them into a detection model or review system



[Safe Task Execution by Agent]





Demo: Exploring AI Control Methods in Agentic AI Systems

